

Remote Controlled STEREO FM RADIO With RDS Using ESP-01

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In this article, we will explore a straightforward method to create an IR remote-controlled FM radio with RDS (radio data system) using the ESP-01 module. We will utilise the Arduino IDE for software development and implementation. Our primary goal is to design an affordable, compact, lightweight, and portable battery-operated FM radio

with RDS message display, complete with IR remote control for user-friendly operation.

While there are several Arduino-based FM radios utilising modules like TEA5767, RDA5807M, and Si4703, they often come with drawbacks such as bulkiness, high power consumption, and excessive cost. For instance, implementing an FM

radio with an Arduino Uno requires a significant amount of power and is not cost-effective. Even the Arduino Nano, though smaller, is relatively expensive and requires a larger number of pins for soldering.

To tackle these limitations, we have chosen the ESP-01 module, which boasts four I/O pins, making it an ideal candidate for our radio. The ESP-01 is easy to program and assemble, requiring minimal expertise. Unlike microcontroller-based designs, which demand additional components and pricey programmers, the ESP-01's programming can be accomplished using an inexpensive USB-to-TTL converter. Furthermore, there are numerous well-established Arduino libraries available for FM radio, IR remote control, and LCD/OLED displays, making software development a breeze.

For I2C communication with the FM radio module we need two I/O pins. For IR remote control we need one I/O pin. With just three I/O pins we can implement an FM radio with

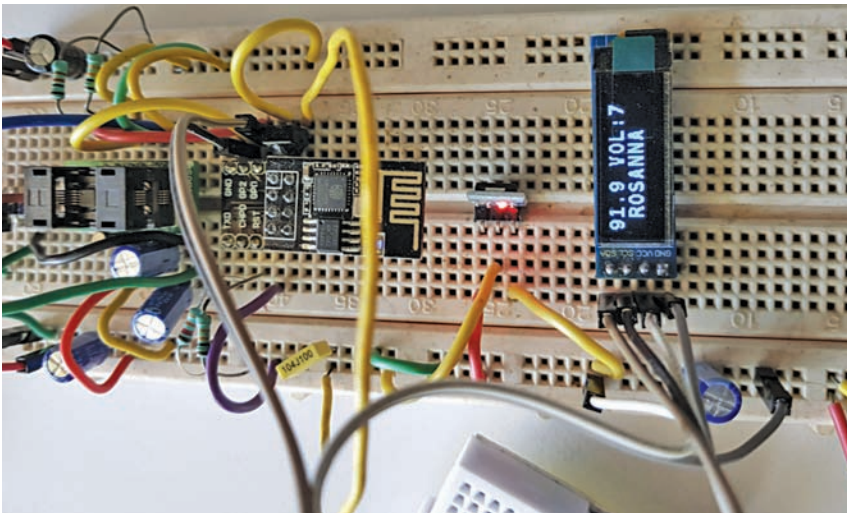


Fig. 1: Author prototype tested on a breadboard

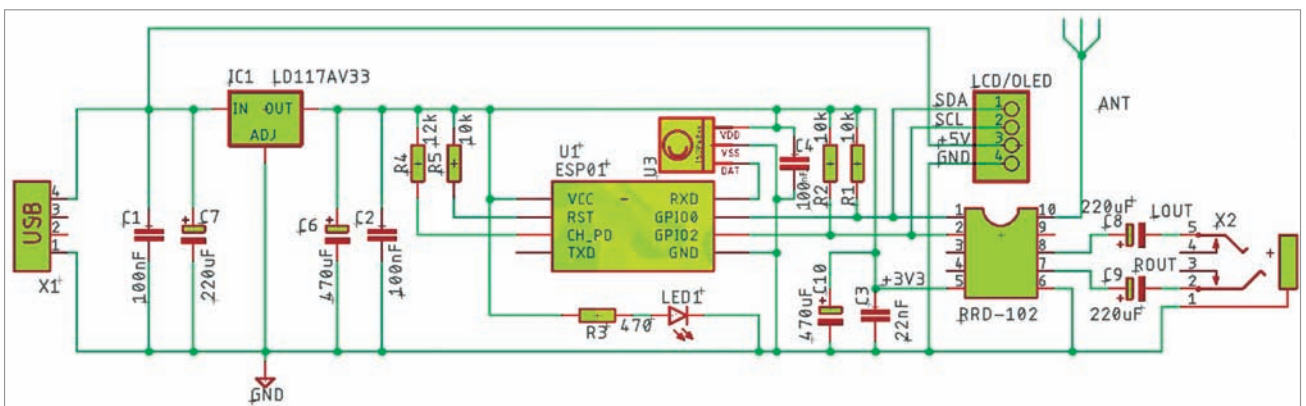


Fig. 2: Circuit diagram